

Conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

23 September 2026

Summary

- Goal: generate the post-spin-up (T, S) state of the Veros ACC channel from the EKE-closure coefficients (c_k, c_ϵ) , skipping the spin-up.
- Pipeline: DINO-Fusion diffusion code adapted to a parameter ensemble; one split runs end to end in under one A100 hour.
- Hold-out RMSE 0.061 / 0.076 / 0.064 / 0.110 / 0.141 K (scattered / band of three rows / centre block of 49 runs / outer ring of 51 runs / top row), against 0.037 / 0.116 / 0.114 / 0.131 / 0.170 K for the nearest training run and 0.155 / 0.197 / 0.197 / 0.258 / 0.470 K for the mean state: the model wins in the four large-gap splits, by a factor 1.8 with the centre held out, and loses only to one-step neighbours. Extrapolating outward from the centre, the error sits in the warm corner (0.3–0.8 K at five runs, cold bias) and stays at 0.04–0.13 K elsewhere. A Wasserstein metric on the stratification profiles gives the same ranking.
- Physical miss: temperature inversions are generated at the rate of the weakly eddying runs in every regime.

1 Data

- 100 Veros ACC runs on a 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$; 100 years, 30-day output; interior grid $30 \times 42 \times 15$ (x, y, z).
- Kept: T and S , last 20 years (241 snapshots per run). $S = 35$ psu everywhere in every run.
- Nearly deterministic end states: within-window std 0.01 K above 1400 m vs 0.05 K between grid neighbours; the bottom level still drifts by 0.45 K over the window.
- One effective parameter: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean T ; the low- c_k/c_ϵ part of the grid saturates.
- Per-level distribution of T (Figure 1): above 400 m a mixture of restoring values and interior, with σ_z set by the meridional gradient, so $\mu_z \pm 3\sigma_z$ is wider than the level and the normalised values stay within $[-0.85, 0.6]$; below 1400 m a warm tail to $x' = +2.7$. Run-to-run signal σ_z^{runs} : 0.7 % of $3\sigma_z$ at the surface, 26 % at the bottom.

2 Method

- Fields $\rightarrow$ 30 channels $(T_{1..15}, S_{1..15})$, normalised per level, land $\rightarrow 0$, zero-padded $42 \times 30 \rightarrow 48 \times 32$:

$$x'_z = \frac{x_z - \mu_z}{3 \max(\sigma_z, 0.05)}. \quad (1)$$

μ_z, σ_z are computed once on all 100 runs and held fixed for every split (a deliberate, mild leakage of 30 scaling constants; no held-out state enters training).

- Denoiser: UNet2DModel (64, 64, 128, 128), 6.9 M parameters. The condition c , the standardised pair $(\log c_k, \log c_\epsilon)$, enters through an MLP added to the timestep embedding.

Distribution of T per level, all 100 runs $\times$ 241 snapshots, water cells (surface at the top). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

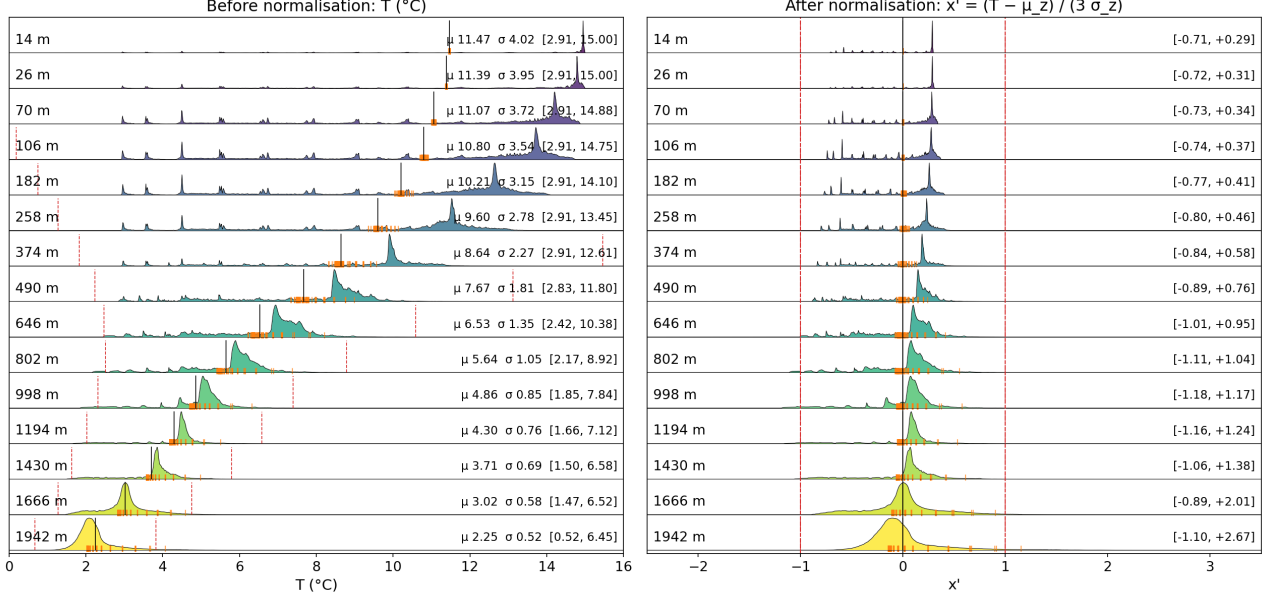


Figure 1: Distribution of T per level over all 100 runs and 241 snapshots (water cells), before (left) and after (right) the per-level normalisation of Eq. (1). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

- DDPM, ϵ -prediction, 1000 steps, squared-cosine schedule, EMA:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c) \right\|^2. \quad (2)$$

20 000 steps, batch 32, AdamW 10^{-4} with cosine decay; 22 min on one A100.

- Sampling: 1000-step DDPM loop with EMA weights; the predicted clean state is clipped at $|x'| \leq 1$; after every step the land and padding cells are reset to zero at the step’s noise level, $\sqrt{1 - \bar{\alpha}_t} z$, as in training. 32 samples per condition.

Differences from the DINO-Fusion code.

- Conditioning: DINO’s UNet is unconditional and is steered at sampling time by constraints; here the UNet receives $(\log c_k, \log c_\epsilon)$ through the timestep embedding, and the data are an ensemble of 100 runs split by run instead of one trajectory.
- Sampling: DINO’s BorderZeroConstraint resets land and padding to exact zeros after every step. In training those cells carry noise like every other, so a solid block of zeros at high noise levels is out of distribution; it is replaced by zeros at the step’s noise level, $\sqrt{1 - \bar{\alpha}_t} z$. On the same trained models this alone takes the hold-out RMSE from 0.152 / 0.135 / 0.136 / 0.173 K to 0.061 / 0.076 / 0.064 / 0.141 K (the four splits of Table 1), with the largest gain at the bottom. DINO’s stratification constraints are not ported yet.
- Everything else is kept: transform chain, 3-std normalisation, DDPM settings, EMA.

3 Evaluation

- Truth: 20-year time-mean state of each held-out run; metrics on water cells.
- Baselines: mean training state; nearest training run in standardised log-parameter space; average of the axis-neighbour training runs on the grid.
- Five hold-out splits, same data file and statistics, different training sets: scattered (10 interior points), band of three rows ($c_k = 0.126, 0.2, 0.3175$; 30 runs; training rows a factor 6.3 apart), centre block (the 7×7 centre of the grid, c_k 0.0315–0.504, c_ϵ 0.22–3.5; 49 runs; training on the

outer rows and columns only), outer ring (the complement: the 51 outer runs held out, training on the centre; extrapolation outward on all four sides), top row ($c_k = 0.8$; 10 runs; extrapolation).

- **Distributional metric:** each state is reduced to its profile of horizontal means over water cells, $\bar{T}_z$; per level, the Wasserstein-1 distance between the 32 generated values and the 241 true snapshots of the window, through the quantile functions,

$$W_1^{(z)} = \int_0^1 |F_{g,z}^{-1}(q) - F_{t,z}^{-1}(q)| dq, \quad W_1 = \sum_z \frac{\Delta z_z}{H} W_1^{(z)}, \quad (3)$$

with Δz_z the level thicknesses ($H = 2080$ m). All 241 snapshots are used: the quantile form accepts unequal sizes and a subsample of 32 would only add noise. A point prediction x (the baselines) gives $W_1^{(z)} = \frac{1}{241} \sum_i |x_z - \bar{T}_{z,i}|$. W_1 rewards a correct spread and does not reward collapse to the mean; the RMSE keeps the horizontal structure.

	scattered 10 runs	band of three 30 runs	centre block 49 runs	outer ring 51 runs	top row 10 runs
diffusion, mean of 32: RMSE (K)	0.061	0.076	0.064	0.110	0.141
diffusion, single sample: RMSE (K)	0.147	0.160	0.153	0.185	0.208
nearest training run: RMSE (K)	0.037	0.116	0.114	0.131	0.170
neighbour average: RMSE (K)	0.020	0.116	0.113	0.131	0.170
mean training state: RMSE (K)	0.155	0.197	0.197	0.258	0.470
diffusion: domain-mean bias (K)	+0.042	+0.020	+0.032	−0.002	−0.016
ensemble spread (K)	0.128	0.132	0.132	0.124	0.132
true within-window spread (K)	0.031	0.031	0.031	0.030	0.032
T inversions, generated	10.2 %	10.9 %	10.4 %	11.0 %	12.3 %
T inversions, truth	7.7 %	1.3 %	4.2 %	6.6 %	0.1 %
max salinity error (psu)	0.013	0.014	0.013	0.013	0.017
W_1 profile, diffusion (K)	0.066	0.075	0.073	0.100	0.132
W_1 profile, neighbour average (K)	0.036	0.095	0.095	0.117	0.142
W_1 profile, nearest training run (K)	0.044	0.095	0.095	0.117	0.142
W_1 profile, mean training state (K)	0.136	0.160	0.166	0.223	0.420

Table 1: Hold-out metrics per split, mean over the held-out runs, water cells, 32 samples per run.

4 Findings

- **Model vs baselines.** Band: beats the nearest run at every level below 26 m and in all three rows (0.063 / 0.074 / 0.091 K vs 0.063 / 0.132 / 0.152 K for $c_k = 0.126$ / 0.2 / 0.3175). Centre block: beats the nearest run at every level below the surface; over the block the error is flat, 0.04–0.12 K (centre 3×3 0.068 K, rim 0.063 K), where the nearest run degrades from 0.10 K at the rim to 0.18 K at the centre; it wins at the 28 runs where the state changes with the parameters (nearest run 0.10–0.39 K) and loses by at most 0.07 K at the 21 runs of the flat regime, where a copied neighbour is already within 0.05 K (Figure 4). Outer ring (training on the centre, extrapolating outward): median 0.061 K, mean 0.110 K; the mean is set by five warm-corner runs (0.27–0.77 K, cold bias down to −0.43 K at the corner), the rest of the ring sits at 0.04–0.13 K; beats the nearest run from 106 to 1666 m and at 24 of 51 runs, the losses again in the flat regime; the nearest run is worse still at the corner (0.92 K) (Figure 5). Top row: beats the nearest run at every level but the surface and the bottom, and beats copying the row below for c_e 0.14–1.4 (0.05–0.26 vs 0.08–0.31 K); loses at the corner (0.41 vs 0.35 K) and in the flat regime ($c_e \geq 3.5$). Scattered: loses to the one-step neighbours (0.061 vs 0.037 / 0.020 K) except between 490 and 650 m (Figures 2–6).
- **Depth structure.** 0.04–0.08 K at every level in the scattered and block splits; the bottom levels are no longer special (0.07 K) except where the warm corner is held out (top row 0.29 K, ring 0.17 K).
- **Why the surface is worst in kelvin.** The per-level scale $3\sigma_z$ is 12 K at the surface and 1.6 K at the bottom, and the sampler’s precision is roughly uniform in normalised units: the surface RMSE is 0.6 % of its scale (0.073 K, scattered), against 1 % at 650 m and 4.5 % at the bottom. The run-to-run surface signal is 0.7 % of the scale (0.085 K), the same size, so the model cannot tell the runs apart there and matches the mean state (0.076 vs 0.073 K); the nearest run wins (0.026 K)

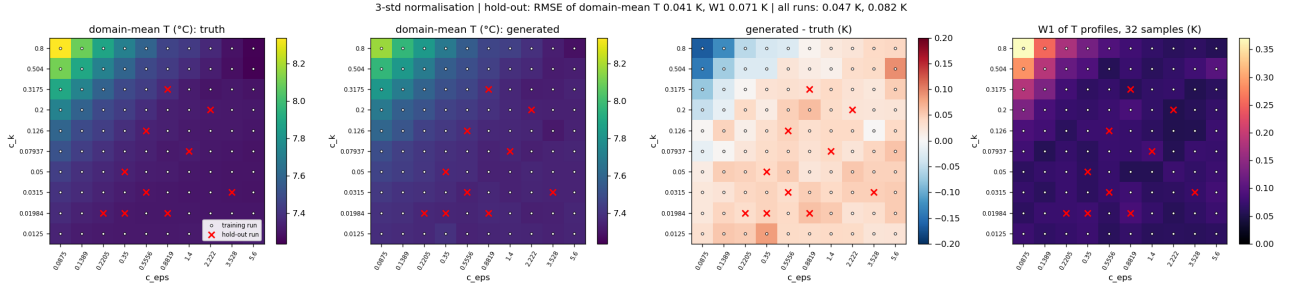
only because neighbouring runs are nearly identical at the surface. What remains is a per-sample offset of the surface mean: W_1 on the horizontal-mean profile is 0.12 K at 14 m, above the 0.07 K cell RMSE of the ensemble mean, so each sample carries a nearly uniform shift of about 0.1 K (0.01 normalised) that the average of 32 mostly removes.

- **The clip is a regulariser, not a range limit.** Same models, sampling only (with DINO’s constraint), scattered / top: clip at $|x'| \leq 1$ gives 0.152 / 0.173 K, at 1.5 0.194 / 0.183 K, at 3 0.302 / 0.277 K, although the final samples barely exceed $|x'| = 1$ (0.1 % of values): clipping the early clean-state estimates keeps the chain on track. Its price is a range limit: 12 % of the held-out top row’s true bottom values lie above $\mu_z + 3\sigma_z$ and cannot be generated; the corner run is the largest error of the top-row and ring splits.
- **Inversions.** 10–12 % of interfaces with $\partial_z T < 0$ in every regime; truth 7.7 % (scattered), 6.6 % (ring), 4.2 % (block), 1.3 % (band), 0.1 % (top). Not learned.
- **W_1 agrees with the RMSE ranking.** Training grid points score like held-out ones (0.083 vs 0.071 K, scattered); from 490 m down the samples beat the point baselines in the band and top splits (0.058 vs 0.115 K at 650 m, band), which pay for the spread they cannot represent.
- **Plumbing.** S within 0.02 psu of 35 with no constraint; land exact; ensemble spread $4\times$ the true within-window spread (Figures 7–11).

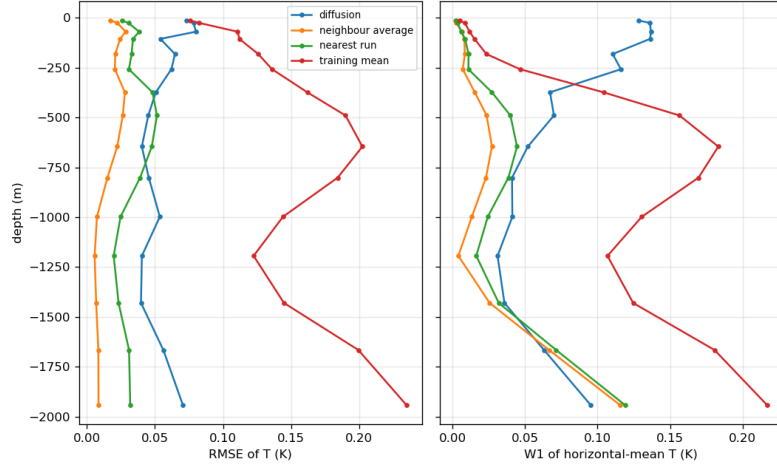
5 Next steps

- Stratification constraint at sampling time (DINO’s isotonic projection on T): targets the inversions.
- Top-row corner: the clip’s range limit is now the dominant extrapolation error; a clip that follows the conditioning (bounds from the nearest training rows) is the next sampling-only test.
- Stronger conditioning: classifier-free guidance; the year as a third condition (bottom drift of 0.45 K inside the window). A second training seed to size the seed noise.
- Evaluate against snapshots as well as the time mean (nearest-snapshot RMSE, spread–skill).
- Then the same pipeline on a global Veros configuration (active S , seasonal cycle, multi-century spin-up).

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`. Training at `2b542aa` (338997d for the block split, 61d0dd2 for the ring); sampling and evaluation at `bfee32f`. Extraction 2.5 min (8 cores), training 17–22 min (one A100), generation of the hold-out and grid sets plus evaluation 14–18 min.

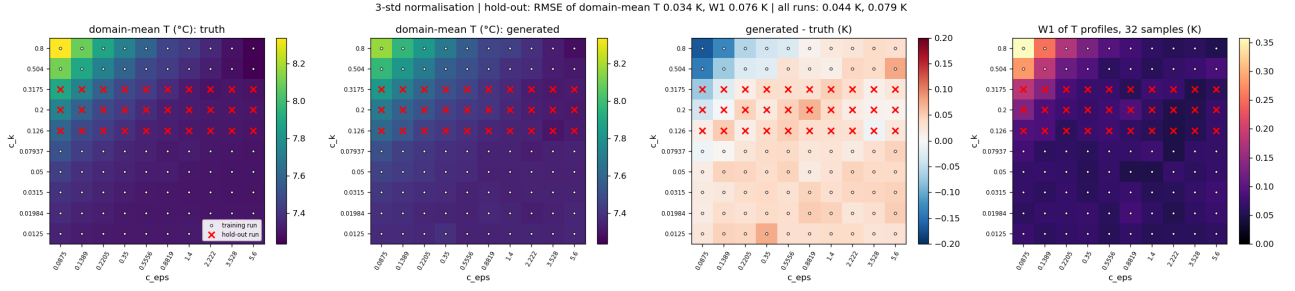


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

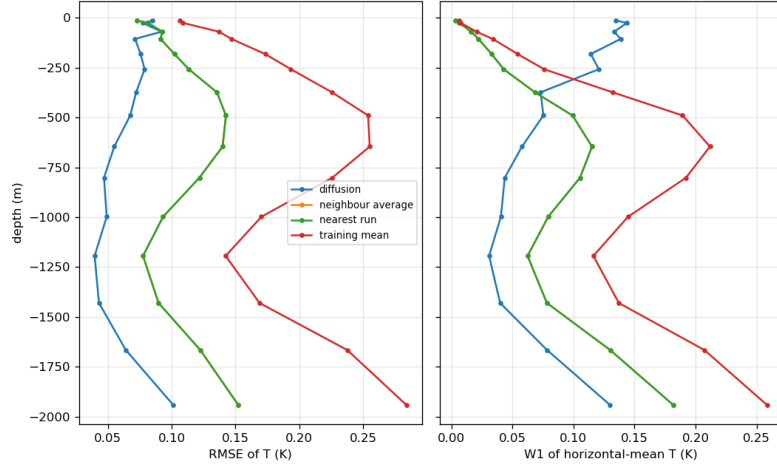


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 2: Scattered split (10 interior held-out runs), 32 samples per run.

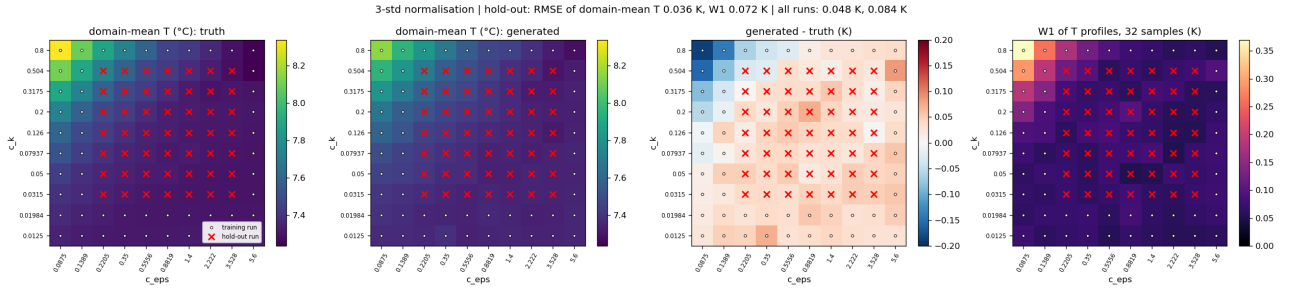


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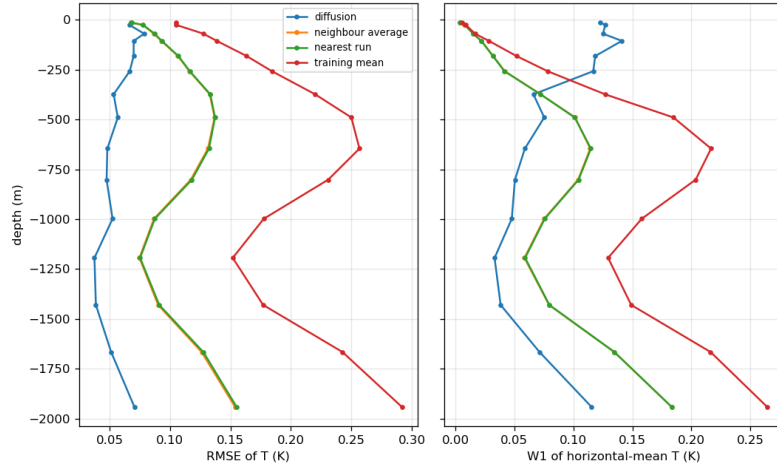


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 3: Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.

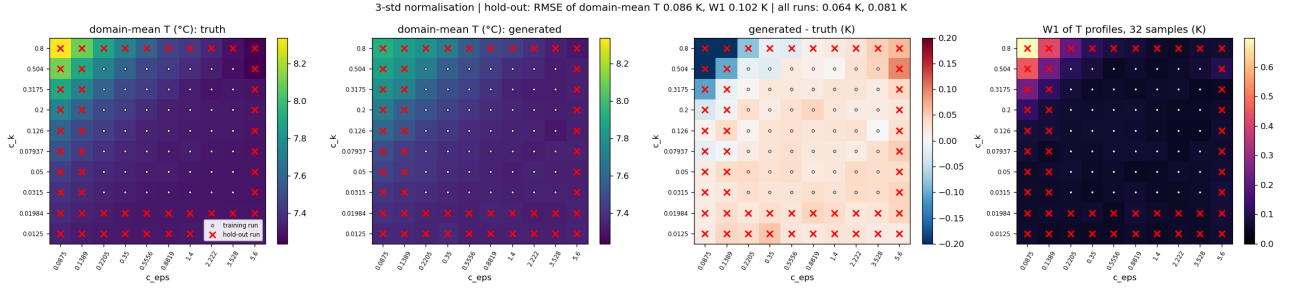


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

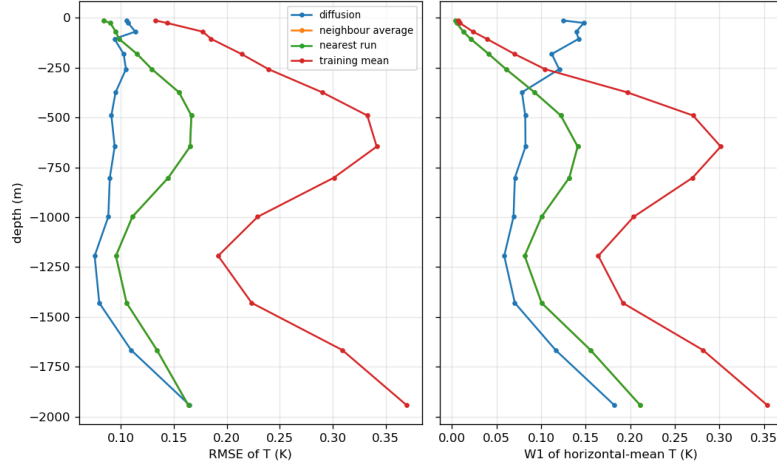


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 4: Centre block (7×7 centre of the grid held out, 49 runs; training on the outer rows and columns), 32 samples per run.

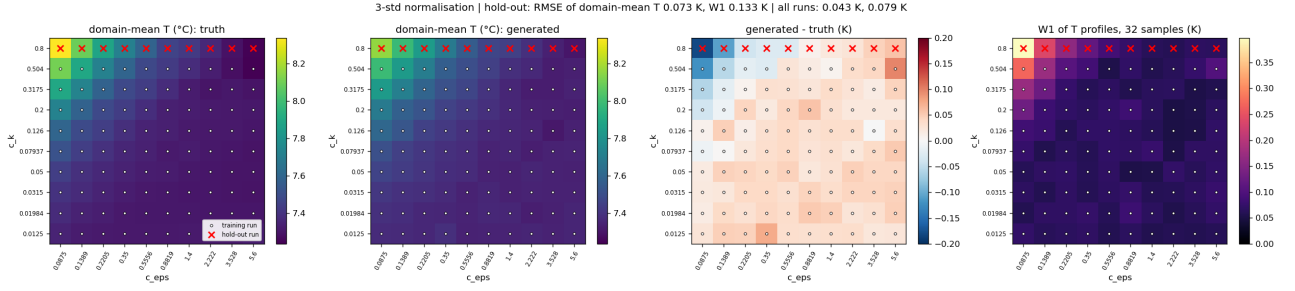


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

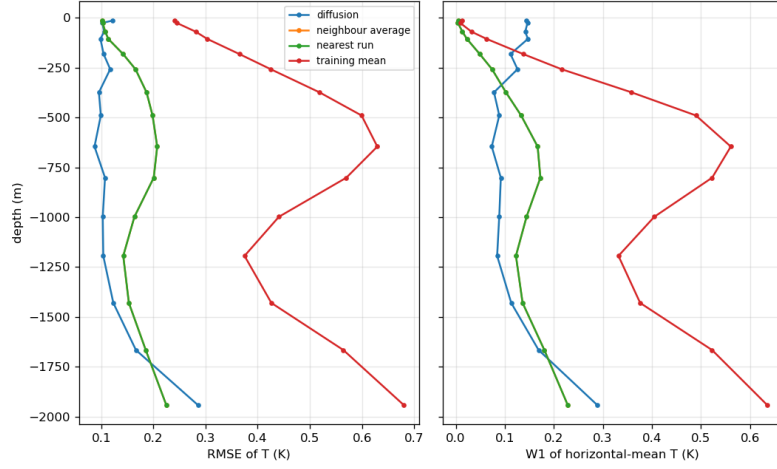


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 5: Outer ring (the 51 outer runs held out; training on the 7×7 centre), 32 samples per run.



(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.



(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 6: Top row ($c_k = 0.8$ held out, extrapolation), 32 samples per run.

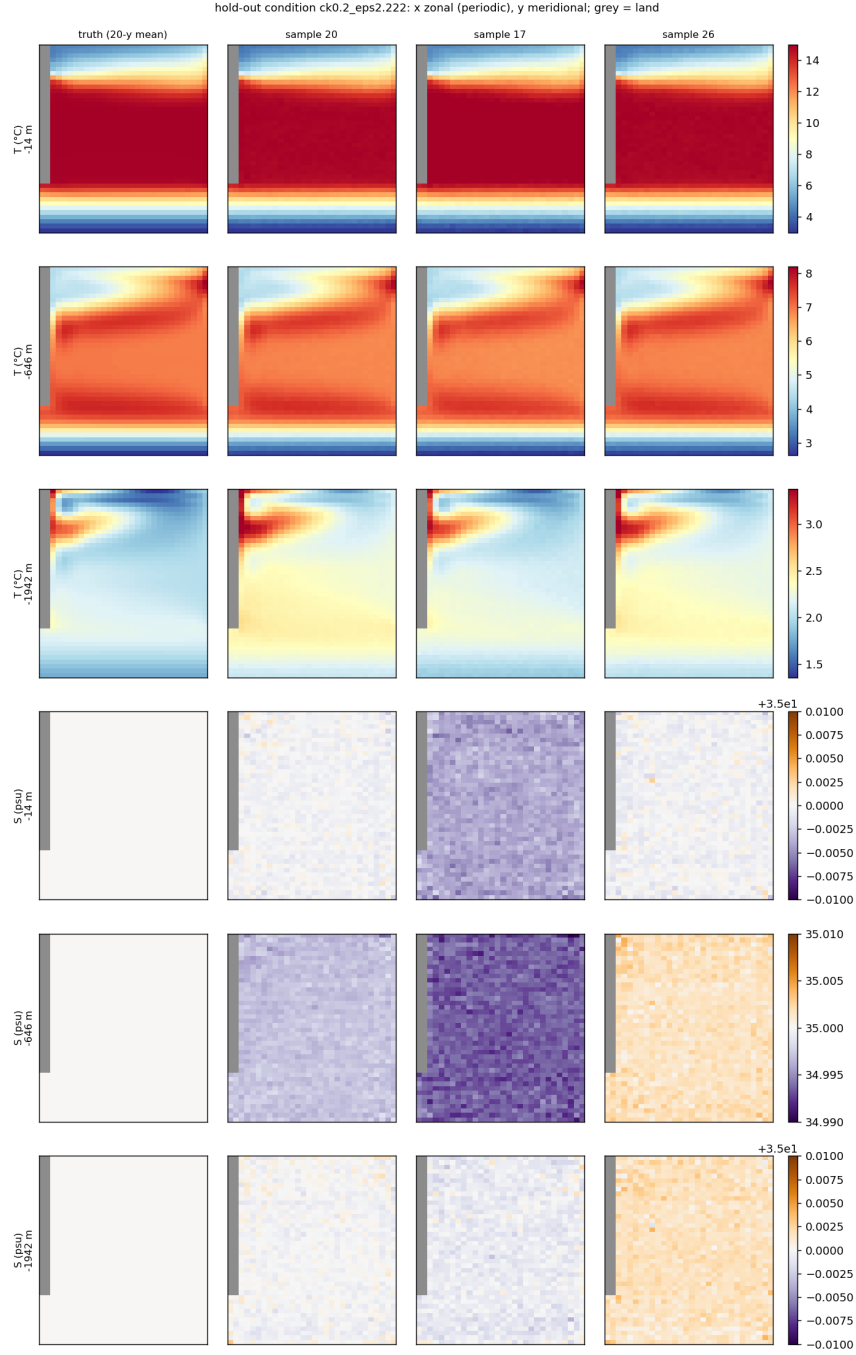


Figure 7: Scattered split, $c_k = 0.2$, $c_\epsilon = 2.222$: true 20-year mean (left) and three random samples at the surface, 650 m and the bottom; T (top three rows), S (bottom three rows). Grey: land.

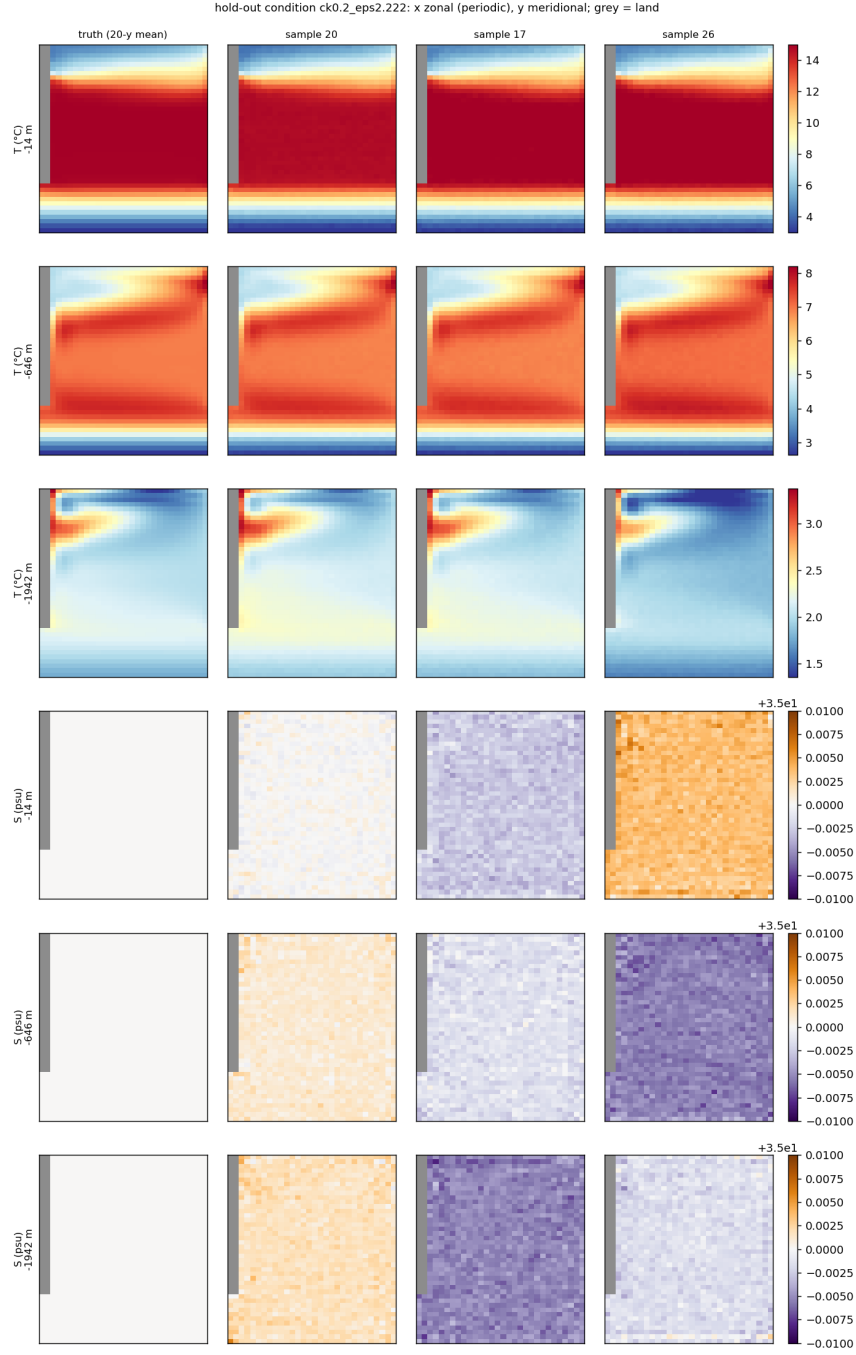


Figure 8: Band-of-three split, same condition ($c_k = 0.2$, $c_{\epsilon} = 2.222$, now in the middle held-out row).

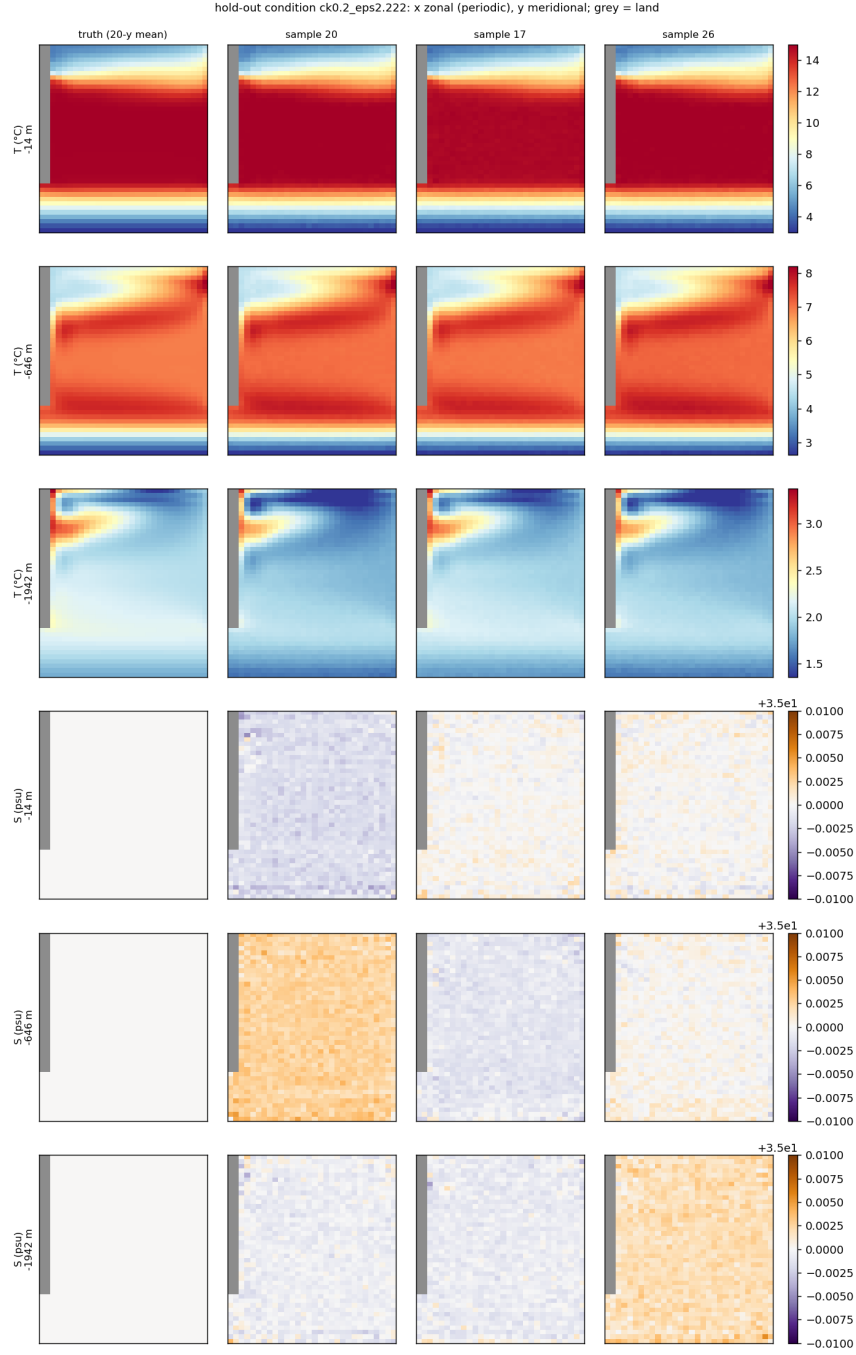


Figure 9: Centre-block split, same condition ($c_k = 0.2$, $c_\epsilon = 2.222$, inside the held-out block).

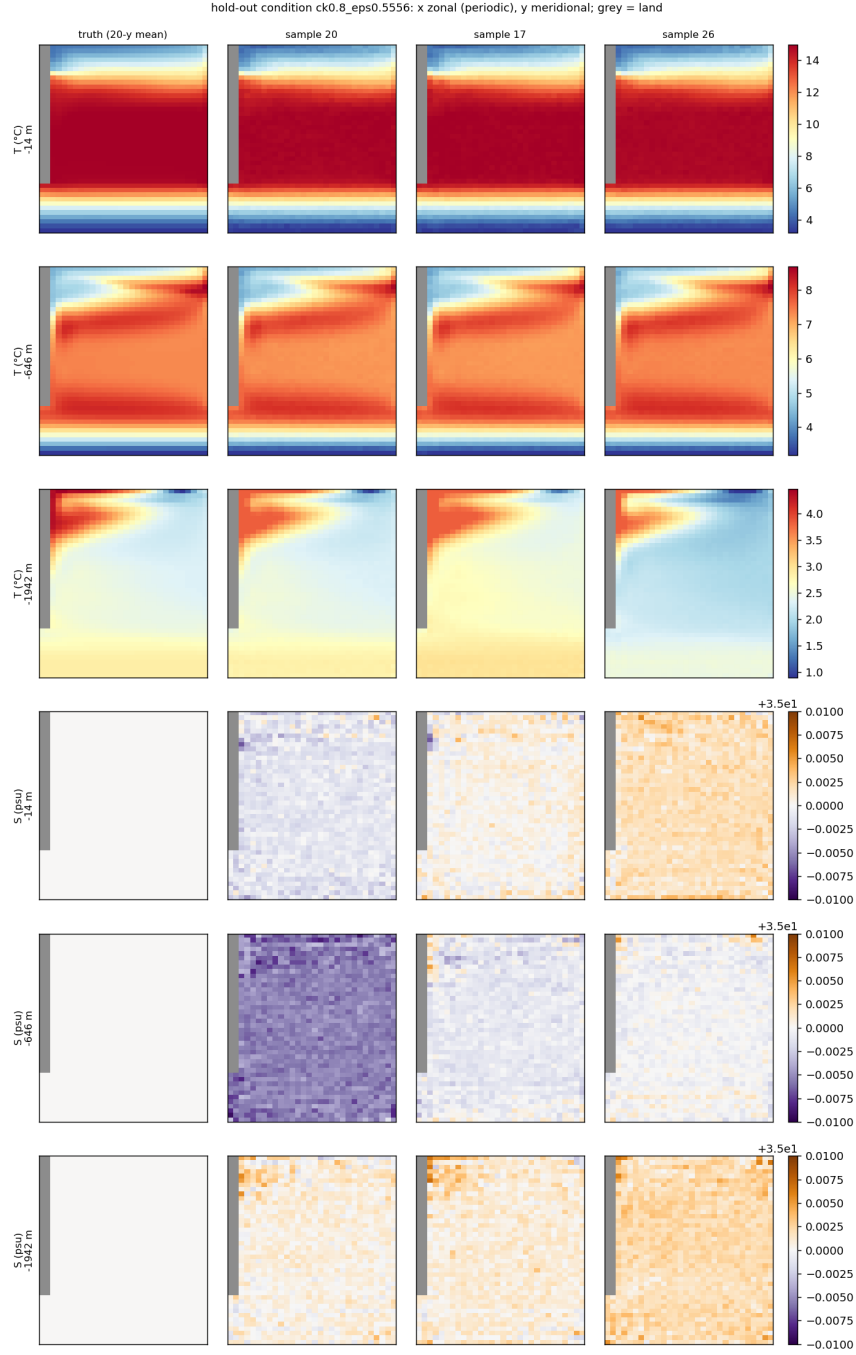


Figure 10: Outer-ring split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (same condition as the top-row split; here the whole outer ring is unseen).

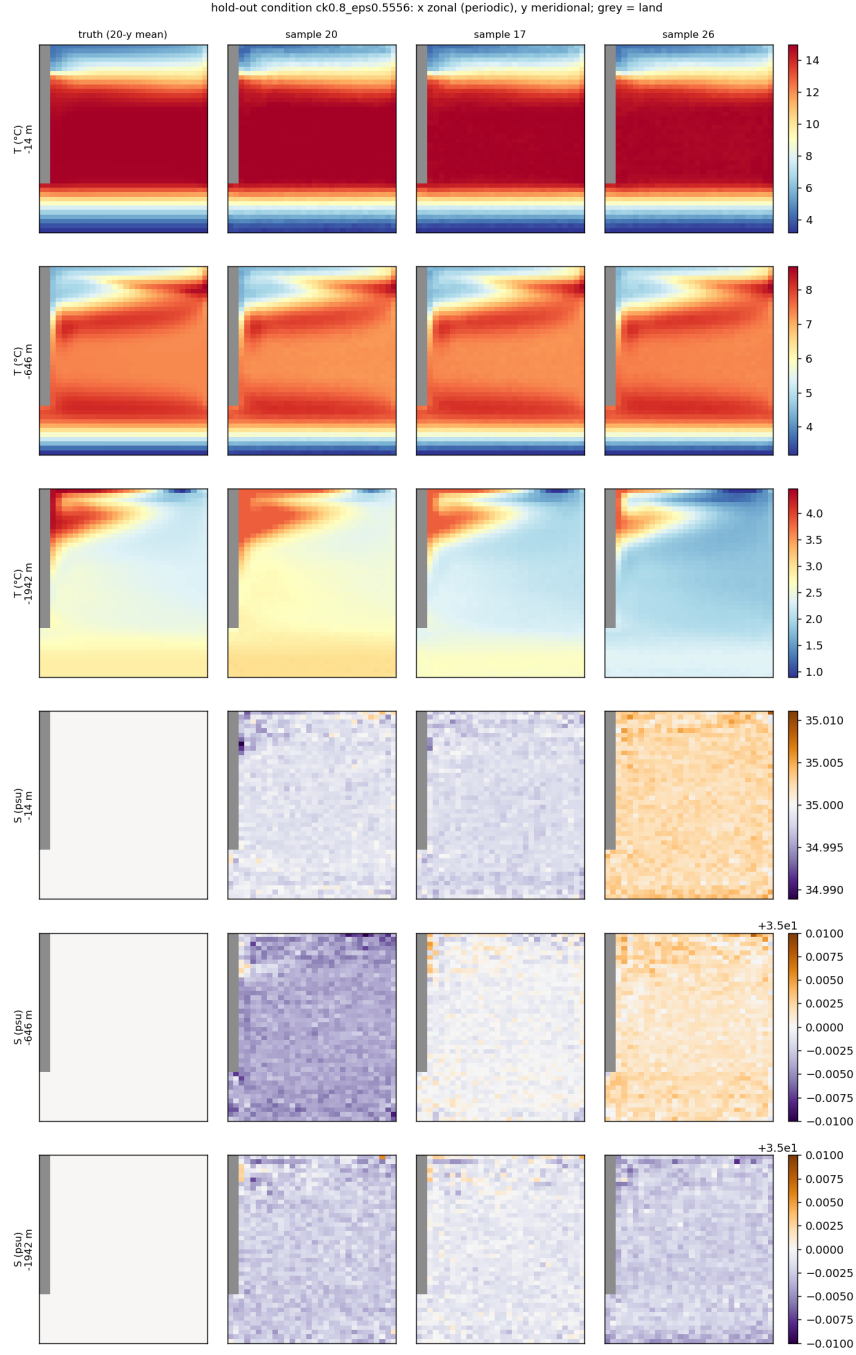


Figure 11: Top-row split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (extrapolation one row above the training grid).